

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
```

```
import pandas as pd

# Load dataset with encoding fix
df = pd.read_csv(
    '/superstore_final_dataset (1).csv',
    encoding='latin1'
)

# View first rows
df.head()
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Post
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
print(df.columns)
print(df.head())
print(df.info())
```

```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
       'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
       'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
       'Product_Name', 'Sales'],
      dtype='object')
Row_ID    Order_ID  Order_Date  Ship_Date  Ship_Mode  Customer_ID \
0         1  CA-2017-152156  8/11/2017  11/11/2017  Second Class  CG-12520
1         2  CA-2017-152156  8/11/2017  11/11/2017  Second Class  CG-12520
2         3  CA-2017-138688  12/6/2017  16/06/2017  Second Class  DV-13045
3         4  US-2016-108966  11/10/2016  18/10/2016  Standard Class  SO-20335
4         5  US-2016-108966  11/10/2016  18/10/2016  Standard Class  SO-20335

Customer_Name  Segment  Country  City  State \
0  Claire Gute  Consumer  United States  Henderson  Kentucky
1  Claire Gute  Consumer  United States  Henderson  Kentucky
2  Darrin Van Huff  Corporate  United States  Los Angeles  California
3  Sean O Donnel  Consumer  United States  Fort Lauderdale  Florida
4  Sean O Donnel  Consumer  United States  Fort Lauderdale  Florida

Postal_Code  Region  Product_ID  Category  Sub_Category \
0  42420.0  South  FUR-B0-10001798  Furniture  Bookcases
1  42420.0  South  FUR-CH-10000454  Furniture  Chairs
2  90036.0  West  OFF-LA-10000240  Office Supplies  Labels
3  33344.0  South  FUR-BA-10000533  Furniture  Tables
```

```
3 33311.0 South FUR-TA-10000577 Furniture Tables
4 33311.0 South OFF-ST-10000760 Office Supplies Storage
```

```
                                Product_Name    Sales
0          Bush Somerset Collection Bookcase  261.9600
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
2  Self-Adhesive Address Labels for Typewriters b...  14.6200
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
4          Eldon Fold N Roll Cart System    22.3680
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 9800 entries, 0 to 9799

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Row_ID	9800 non-null	int64
1	Order_ID	9800 non-null	object
2	Order_Date	9800 non-null	object
3	Ship_Date	9800 non-null	object
4	Ship_Mode	9800 non-null	object
5	Customer_ID	9800 non-null	object
6	Customer_Name	9800 non-null	object
7	Segment	9800 non-null	object
8	Country	9800 non-null	object
9	City	9800 non-null	object
10	State	9800 non-null	object
11	Postal_Code	9789 non-null	float64
12	Region	9800 non-null	object
13	Product_ID	9800 non-null	object
14	Category	9800 non-null	object
15	Sub_Category	9800 non-null	object
16	Product_Name	9800 non-null	object
17	Sales	9800 non-null	float64

dtypes: float64(2), int64(1), object(15)

memory usage: 1.3+ MB

None

```
import os
```

```
print(os.getcwd())
print(os.listdir())
```

```
/content
['.config', 'drive', 'sample_data']
```

```
import os
```

```
print(os.listdir('/'))
```

```
['root', 'etc', 'sbin', 'run', 'media', 'proc', 'lib32', 'srv', 'sys', 'bin', 'dev', 'home', 'lib64', 'var', 'libx32', 'usr',
```

```
import pandas as pd
```

```
df = pd.read_csv(
    '/content/sale/superstore_final_dataset (1).csv',
    encoding='ISO-8859-1'
)
```

```
print(df.head())
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	\
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	

	Customer_Name	Segment	Country	City	State	\
0	Claire Gute	Consumer	United States	Henderson	Kentucky	
1	Claire Gute	Consumer	United States	Henderson	Kentucky	
2	Darrin Van Huff	Corporate	United States	Los Angeles	California	
3	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	
4	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	

	Postal_Code	Region	Product_ID	Category	Sub_Category	\
0	42420.0	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420.0	South	FUR-CH-10000454	Furniture	Chairs	
2	90036.0	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311.0	South	FUR-TA-10000577	Furniture	Tables	
4	33311.0	South	OFF-ST-10000760	Office Supplies	Storage	

```

      Product_Name      Sales
0      Bush Somerset Collection Bookcase 261.9600
1  Hon Deluxe Fabric Upholstered Stacking Chairs,... 731.9400
2  Self-Adhesive Address Labels for Typewriters b... 14.6200
3      Bretford CR4500 Series Slim Rectangular Table 957.5775
4      Eldon Fold N Roll Cart System 22.3680
```

```
print(df.columns)

print(df.info())

print(df.isnull().sum())
```

```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
      'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
      'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
      'Product_Name', 'Sales'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row_ID                9800 non-null  int64
1   Order_ID              9800 non-null  object
2   Order_Date            9800 non-null  object
3   Ship_Date             9800 non-null  object
4   Ship_Mode             9800 non-null  object
5   Customer_ID           9800 non-null  object
6   Customer_Name         9800 non-null  object
7   Segment               9800 non-null  object
8   Country               9800 non-null  object
9   City                  9800 non-null  object
10  State                 9800 non-null  object
11  Postal_Code           9789 non-null  float64
12  Region                9800 non-null  object
13  Product_ID            9800 non-null  object
14  Category              9800 non-null  object
15  Sub_Category          9800 non-null  object
16  Product_Name          9800 non-null  object
17  Sales                 9800 non-null  float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
None
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 11
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
dtype: int64
```

```
import pandas as pd

# Load dataset
df = pd.read_csv(
    '/content/sale/superstore_final_dataset (1).csv',
    encoding='latin1'
)

# Show first 5 rows
print(df.head())
```

```

      Row_ID      Order_ID  Order_Date  Ship_Date      Ship_Mode  Customer_ID \
0         1  CA-2017-152156   8/11/2017  11/11/2017    Second Class    CG-12520
1         2  CA-2017-152156   8/11/2017  11/11/2017    Second Class    CG-12520
2         3  CA-2017-138688  12/6/2017   16/06/2017    Second Class    DV-13045
```

```

3      4 US-2016-108966 11/10/2016 18/10/2016 Standard Class SO-20335
4      5 US-2016-108966 11/10/2016 18/10/2016 Standard Class SO-20335

```

```

      Customer_Name  Segment      Country      City      State \
0    Claire Gute    Consumer  United States      Henderson  Kentucky
1    Claire Gute    Consumer  United States      Henderson  Kentucky
2  Darrin Van Huff  Corporate  United States      Los Angeles  California
3    Sean O Donnel  Consumer  United States  Fort Lauderdale  Florida
4    Sean O Donnel  Consumer  United States  Fort Lauderdale  Florida

```

```

      Postal_Code Region      Product_ID      Category Sub_Category \
0    42420.0 South  FUR-BO-10001798      Furniture  Bookcases
1    42420.0 South  FUR-CH-10000454      Furniture  Chairs
2    90036.0 West  OFF-LA-10000240  Office Supplies  Labels
3    33311.0 South  FUR-TA-10000577      Furniture  Tables
4    33311.0 South  OFF-ST-10000760  Office Supplies  Storage

```

```

      Product_Name      Sales
0    Bush Somerset Collection Bookcase  261.9600
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
2  Self-Adhesive Address Labels for Typewriters b...  14.6200
3    Bretford CR4500 Series Slim Rectangular Table  957.5775
4    Eldon Fold N Roll Cart System  22.3680

```

```

# Number of rows and columns
print(df.shape)

```

```

# Column names
print(df.columns)

```

```

# Dataset information
print(df.info())

```

```

(9800, 18)
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
      'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
      'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
      'Product_Name', 'Sales'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Row_ID      9800 non-null   int64
1   Order_ID    9800 non-null   object
2   Order_Date  9800 non-null   object
3   Ship_Date   9800 non-null   object
4   Ship_Mode   9800 non-null   object
5   Customer_ID 9800 non-null   object
6   Customer_Name 9800 non-null   object
7   Segment     9800 non-null   object
8   Country     9800 non-null   object
9   City        9800 non-null   object
10  State       9800 non-null   object
11  Postal_Code  9789 non-null   float64
12  Region      9800 non-null   object
13  Product_ID  9800 non-null   object
14  Category    9800 non-null   object
15  Sub_Category 9800 non-null   object
16  Product_Name 9800 non-null   object
17  Sales       9800 non-null   float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
None

```

```

print(df.isnull().sum())

```

```

Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID  0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 11
Region      0

```

```
Product_ID      0
Category        0
Sub_Category    0
Product_Name    0
Sales           0
dtype: int64
```

```
df.drop_duplicates(inplace=True)
```

```
print("Duplicates removed")
```

```
Duplicates removed
```

```
print(df.columns)
```

```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
       'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
       'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
       'Product_Name', 'Sales'],
      dtype='object')
```

```
# Convert Order_Date into datetime
df['Order_Date'] = pd.to_datetime(
    df['Order_Date'],
    dayfirst=True
)

print(df['Order_Date'].head())
```

```
0    2017-11-08
1    2017-11-08
2    2017-06-12
3    2016-10-11
4    2016-10-11
Name: Order_Date, dtype: datetime64[ns]
```

```
# Create Month column
df['Month'] = df['Order_Date'].dt.month

# Create Year column
df['Year'] = df['Order_Date'].dt.year

# Check output
print(df[['Order_Date', 'Month', 'Year']].head())
```

```
   Order_Date  Month  Year
0  2017-11-08     11  2017
1  2017-11-08     11  2017
2  2017-06-12      6  2017
3  2016-10-11     10  2016
4  2016-10-11     10  2016
```

```
print(df.describe())
```

```

      count      Row_ID      Order_Date      Postal_Code      Sales \
mean    4900.500000    2017-05-01 05:13:51.673469440    55273.322403    230.769059
min       1.000000      2015-01-03 00:00:00      1040.000000      0.444000
25%    2450.750000      2016-05-24 00:00:00    23223.000000     17.248000
50%    4900.500000      2017-06-26 00:00:00    58103.000000     54.490000
75%    7350.250000      2018-05-15 00:00:00    90008.000000    210.605000
max    9800.000000      2018-12-30 00:00:00   99301.000000   22638.480000
std    2829.160653                NaN    32041.223413    626.651875

      count      Month      Year
mean     7.818469    2016.724184
min       1.000000    2015.000000
25%       5.000000    2016.000000
50%       9.000000    2017.000000
75%      11.000000    2018.000000
max      12.000000    2018.000000
std       3.281905      1.123984
```

```
print(df.isnull().sum())
```

```
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 11
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
Month       0
Year        0
dtype: int64
```

```
# Fill missing Postal_Code values with 0
df['Postal_Code'] = df['Postal_Code'].fillna(0)

print(df.isnull().sum())
```

```
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 0
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
Month       0
Year        0
dtype: int64
```

```
df.drop_duplicates(inplace=True)

print("Duplicates removed successfully")

Duplicates removed successfully
```

```
total_sales = df['Sales'].sum()

print("Total Revenue:", total_sales)

Total Revenue: 2261536.7827000003
```

```
average_sales = df['Sales'].mean()

print("Average Sales:", average_sales)

Average Sales: 230.7690594591837
```

```
max_sale = df['Sales'].max()

print("Maximum Sale:", max_sale)

Maximum Sale: 22638.48
```

```
lowest_sale = df['Sales'].min()

print(lowest_sale)
```

```
0.444
```

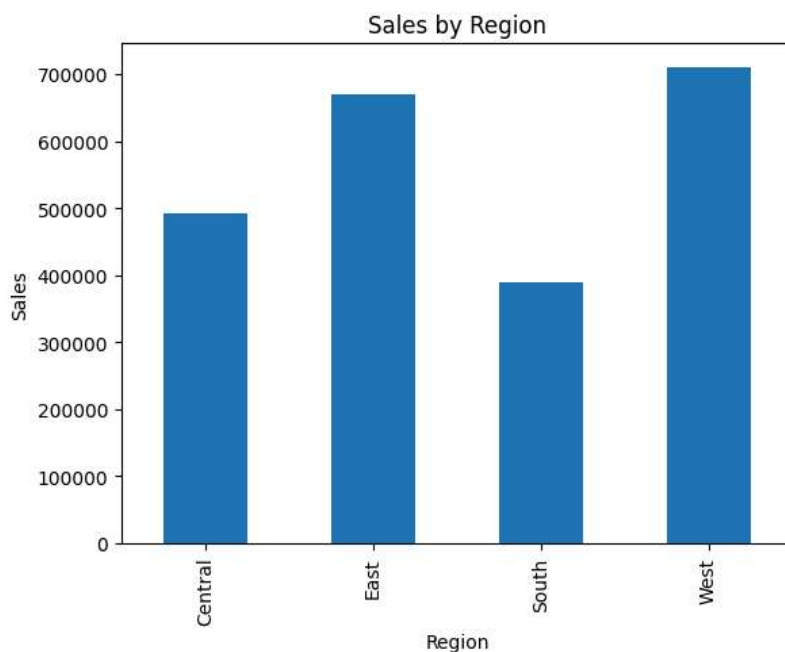
```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
region_sales = df.groupby('Region')['Sales'].sum()

region_sales.plot(kind='bar')

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.show()
```



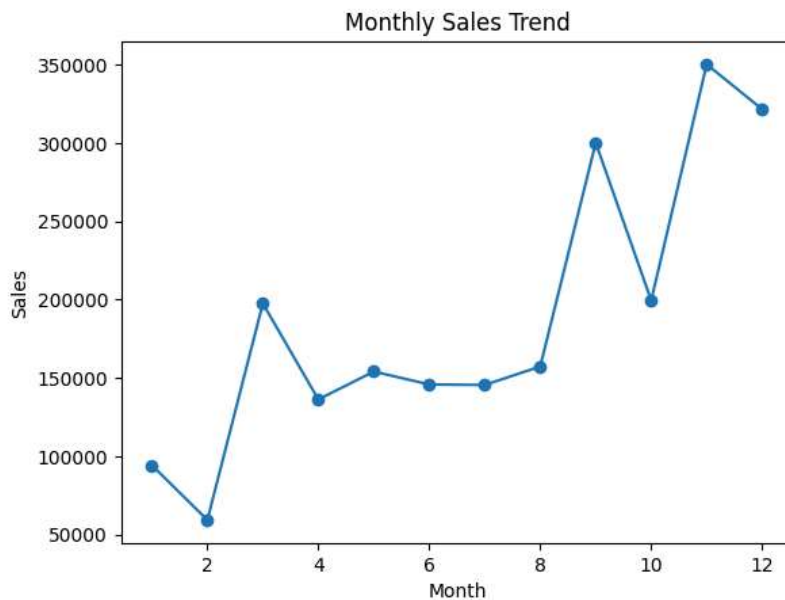
Insight: The West region generated the highest sales revenue compared to other regions.

```
monthly_sales = df.groupby('Month')['Sales'].sum()

monthly_sales.plot(marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

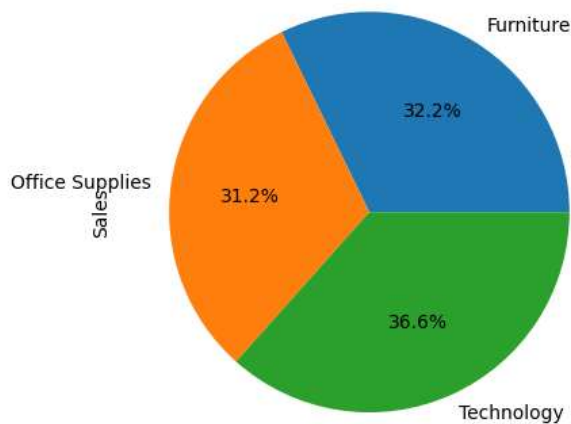
plt.show()
```



```
category_sales = df.groupby('Category')['Sales'].sum()

category_sales.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.show()
```



```
top_products = df.groupby('Product_Name')['Sales'].sum()

top_products = top_products.sort_values(
    ascending=False
).head(10)

print(top_products)
```

Product_Name	Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24 Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

Name: Sales, dtype: float64

1. Increase marketing in high-performing regions.
2. Promote best-selling products more aggressively.
3. Improve pricing strategy for weak-performing products.
4. Prepare inventory before peak sales months.
5. Create combo offers to increase sales.

This project analyzed Superstore sales data using Python and visualization techniques. The analysis identified regional sales trends, top-performing products, and monthly sales patterns. The findings can help Alfido Tech improve marketing strategy, product performance, and inventory planning using data-driven insights.